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United States Patent Application Publication

20250276157

Kind Code

A1

Publication Date

September 04, 2025

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GUIDEWIRE CLAMPING DEVICE AND COLLET

Abstract

A guidewire clamping device system that includes a handle with a body that defines a lumen that extends from a proximal end of the handle to a distal end of the handle. The distal end of the handle comprising a coupling member. The system further includes a collar that defines a lumen that extends from a proximal end to a distal end of the collar. The collar is configured to couple to the distal end of the handle. The system further includes a collet that includes a body with a tubular section with a lumen, and a plurality of ramps that are flared outward from the tubular section in an unconstrained state. The collet is disposed within a lumen of the handle and the collar. The collar is coupled to the handle which biases the plurality of ramps of the collet inward to a constrained state.

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Family ID: 96881685

Appl. No.: 19/066721

Filed: February 28, 2025

Related U.S. Application Data

us-provisional-application US 63560315 20240301

Publication Classification

Int. Cl.: A61M25/09 (20060101)

U.S. Cl.:

CPC A61M25/09041 (20130101); A61M2025/09125 (20130101)

Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/560,315, filed on Mar. 1, 2024, and titled, “Guidewire Clamping Device and Collet,” which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to the field of guidewire clamping devices. More particularly, some embodiments relate to a collet used in guidewire clamping devices to clamp and secure a guidewire.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The written disclosure herein describes illustrative embodiments that are non-limiting and non-exhaustive. Reference is made to certain of such illustrative embodiments that are depicted in the figures, in which:

[0004] FIG. 1 illustrates a hemostasis torque assembly utilizing a guidewire clamping device according to one embodiment of the present disclosure.

[0005] FIG. 2A illustrates a guidewire clamping device according to one embodiment of the present disclosure.

[0006] FIG. 2B illustrates the guidewire clamping device of FIG. 2A clamping a guidewire.

[0007] FIG. 3 illustrates an exploded view of a guidewire clamping device according to one embodiment of the present disclosure.

[0008] FIG. 4A illustrates a perspective view of a handle of a guidewire clamping device according to one embodiment of the present disclosure.

[0009] FIG. 4B illustrates another perspective view of the handle of FIG. 4A.

[0010] FIG. 5A illustrates a perspective view of a collar of a guidewire clamping device according to one embodiment of the present disclosure.

[0011] FIG. 5B illustrates another perspective view of the collar of FIG. 5A.

[0012] FIG. 6A illustrates a perspective view of a collet of a guidewire clamping device according to one embodiment of the present disclosure.

[0013] FIG. 6B illustrates a side view of the collet of FIG. 6A.

[0014] FIG. 6C illustrates an end view of the collet of FIG. 6A.

[0015] FIG. 6D illustrates another perspective view of the collet of FIG. 6A.

[0016] FIG. 7A illustrates a cross-sectional view of a guidewire clamping device with a collet in an unconstrained configuration, according to one embodiment of the present disclosure.

[0017] FIG. 7B illustrates a cross-sectional view of the guidewire clamping device of FIG. 7A with the collet in a partially constrained configuration.

[0018] FIG. 7C illustrates a cross-sectional view of the guidewire clamping device of FIG. 7A with the collect in a constrained configuration.

DETAILED DESCRIPTION

[0019] This disclosure describes a guidewire clamping device for securing a guidewire. The guidewire clamping device may be used in conjunction with another medical device, such as a hemostasis valve. In some instances, when access to a cardiovascular system of a patient is sought, the guidewire clamping device helps provide control to the practitioner of inserted guidewires, such as rotating the guidewire to control the position of a distal tip of the guidewire. Various medical procedures require manipulation of the guidewire. In some procedures, guidewires are designed to navigate blood vessels to reach a target vessel segment. Once at the target vessel segment, the

guidewire may act as a guide for catheters or other devices. Thus, the position of a guidewire may further facilitate proper placement of medical devices. In some procedures, a distal tip of the guidewire must be placed in the desired position at the target vessel segment to properly place a medical tool.

[0020] In some operations, after a medical device is guided to a treatment site, the guidewire is removed while the medical device is used. A physician may need to later reinsert the guidewire to guide additional medical tools to the treatment site. However, reinsertion of the guidewire may alter the axial displacement and/or rotational angle of the guidewire.

[0021] Further, the positional relationship between the end of a guidewire and the end of a catheter may also be related to storage and shipment of a catheter system. For example, to assist with insertion into a blood vessel, a catheter may have a curved end. For shipment, the catheter may be coupled to the hemostasis valve and a guidewire inserted into the catheter. However, if the guidewire extends to the curved end of the catheter, the guidewire may exert a force on the catheter that causes the curved end to lose its form. Thus, in some instances, a guidewire may be placed offset from the curved end during shipping and storage.

[0022] The guidewire clamping device system that works in conjunction with other medical devices may include a handle with a gripping member. The practitioner is able to grip the handle of the guidewire clamping device because the handle is bigger than the guidewire. The gripping member may have a textured surface to increase friction for better gripping. The ability of the practitioner to grip the handle enables the practitioner to rotate the guidewire when the guidewire clamping device clamps the guidewire.

[0023] The guidewire clamping device system may further include a collar with a mating mechanism configured to selectively couple to the handle. A collet within the collar may selectively provide a clamping force to maintain a position of the guidewire extending through the collet. The handle may couple to the collar and selectively alter the clamping force of the collet based on the axial position of the handle relative to the collar. The guidewire may be rotated by the practitioner by rotating the handle when the guidewire clamping device clamps the guidewire. When the guidewire clamping device does not clamp the guidewire, the practitioner may advance and retract the guidewire through a lumen of the guidewire clamping device.

[0024] The phrase “coupled to” is broad enough to refer to any suitable coupling or other form of interaction between two or more entities, including mechanical, fluidic, and thermal interaction. Thus, two components may be coupled to each other even though they are not in direct contact with each other. The phrase “attached to” refers to interaction between two or more entities which are in direct contact with each other and/or are separated from each other only by a fastener of any suitable variety (e.g., mounting hardware or an adhesive).

[0025] Axial displacement refers to movement in a longitudinal direction of an elongated member. Rotational displacement refers to a change in angle of an elongated member about a longitudinal axis.

[0026] The terms “proximal” and “distal” are opposite directional terms. The distal end of a device or component is the end of the component that is furthest from the physician during ordinary use. The proximal end refers to the opposite end, or the end nearest the physician during ordinary use.

[0027] The components of the embodiments as generally described and illustrated in the figures herein can be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of various embodiments, as represented in the figures, is not intended to limit the scope of the present disclosure, but is merely representative of various embodiments. While various aspects of the embodiments are presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

[0028] FIG. 1 illustrates a perspective view of a hemostasis torque assembly **100** according to one embodiment of the present disclosure. The hemostasis torque assembly **100** may facilitate manipulation of a guidewire **102** while limiting fluid loss. The hemostasis torque assembly **100**

illustrated in FIG. 1 includes a guidewire clamping device **150** and a hemostasis valve **110**. The guidewire clamping device **150** enables a practitioner to control and manipulate an inserted guidewire. In some embodiments, the guidewire clamping device **150** and the hemostasis torque assembly **100** may be selectively coupled to each other. In some embodiments, the guidewire clamping device **150** may be used in connection with the hemostasis valve **110**, but those components may only interact via other components (e.g., a guidewire) with no direct attachment or interaction between the hemostasis valve and the guidewire clamping device **150**. Still further, as discussed above, the guidewire clamping device **150** may be used in conjunction with a variety of different medical devices and the present disclosure is not limited to using the guidewire clamping device **150** with the hemostasis valve **110**.

[0029] The guidewire clamping device **150** is configured to selectively secure, engage, or grip a guidewire **102** and facilitate displacement, such as translation or rotation, of the guidewire **102** by manipulation of the guidewire clamping device **150**. For example, as shown, a practitioner may decrease a diameter of an opening of the guidewire clamping device **150** through which the guidewire **102** is placed. When the diameter of the opening is equivalent to the diameter of the guidewire **102**, the guidewire clamping device **150** physically engages with the guidewire **102** such that the position of the guidewire clamping device **150** relative to the guidewire **102** is fixed. Components configured for gripping or securing the guidewire **102** are discussed in more detail below.

[0030] The guidewire clamping device **150** may permit the practitioner to have enhanced or greater rotational control and axial displacement control of the guidewire **102**. For instance, the guidewire clamping device **150** has a greater diameter than the guidewire **102**, thereby providing a larger surface to grip while the practitioner is manipulating the guidewire **102**. Also, when the guidewire clamping device **150** is secured to the guidewire **102**, the coupling of the guidewire clamping device **150** and the hemostasis valve **110** (e.g., directly or via the guidewire **102**) may prevent or limit axial displacement.

[0031] The guidewire clamping device **150** may comprise a handle **160**, a collar **170**, and a collet **180** (not shown in FIG. 1, see FIG. 3). The collet **180** may have a segmented portion. A diameter of the segmented portion may be adjusted by applying or removing a compressing force. The handle **160** and the collar **170** may provide the compressing force. For example, when the handle **160** is axially displaced relative to the collar **170**, the compressing force may be altered. In some embodiments, the axial displacement may be controlled via a coupling member, e.g., a threaded interface between the handle **160** and the collar **170**. In some embodiments, gripping elements **176** may assist in rotating the collar **170** to adjust the axial displacement between the handle **160** and the collar **170**. In some embodiments, the coupling member may facilitate a direct push or pull of the handle **160** in the axial direction to alter the compression force. In some embodiments, the compressing force may be automatically controlled via an actuator.

[0032] The hemostasis valve **110** is configured to prevent fluid loss while allowing medical devices to enter a valve lumen. The hemostasis valve **110** may include an intravascular access connector **120**, a body **130**, and a compression nut **140**.

[0033] The intravascular access connector **120** may be located at a distal end of the body **130**. The intravascular access connector **120** may couple to various medical devices such as a catheter **104** or an introducer. The intravascular access connector **120** may include gripping elements **124** to assist in attaching the intravascular access connector **120** to a medical device.

[0034] The body **130** may include a valve lumen to provide a passage from a distal end to a proximal end. Medical devices, tissue samples, or fluids may be distributed through the valve lumen. A seal within the compression nut **140** may provide a hemostatic entrance to the valve lumen. A sidearm **132** may be coupled to the body **130** and provide a secondary access passage to the valve lumen.

[0035] The hemostasis torque assembly **100** may comprise a seal actuator, configured to selectively

seal the valve lumen. The seal actuator may be configured to alter the diameter of a sealable opening in the seal, allowing a practitioner to obstruct fluid passage through the valve lumen. The seal actuator may facilitate sealing of the valve lumen around an elongate instrument such as the guidewire **102**. In the illustrated embodiment, the compression nut **140** is an exemplary embodiment of a seal actuator. In the illustrated embodiment, a practitioner may use the compression nut **140** to adjust a seal. For example, the diameter of an opening in the seal may change based on a rotational movement of the compression nut **140**. The compression nut **140** may include gripping elements **144** to assist in rotating the compression nut **140**. The gripping elements (**124**, **144**, **154**) in the illustrated embodiment include a plurality of ridges. However, the gripping elements (**124**, **144**, **154**) may comprise other structures, such as divots, high-friction surfaces, and/or slots.

[0036] In some embodiments, the hemostasis valve **110** may provide resistance to prevent or limit unintentional rotational movement of an inserted guidewire **102**. In some embodiments, the collar **170** and the compression nut **140** may comprise indicators that indicate the rotational position of the guidewire clamping device **150** relative to the hemostasis valve **110**.

[0037] FIG. 2A illustrates a detailed perspective view of the guidewire clamping device **150** without a guidewire disposed within the guidewire clamping device **150**. FIG. 2B illustrates a perspective view of the guidewire clamping device **150** with a guidewire **102** extending through the guidewire clamping device **150**. The guidewire **102** may be advanced and slid through the guidewire clamping device **150** by a practitioner until the guidewire **102** reaches the target location in the patient. The guidewire clamping device **150** is also configured to clamp and secure the guidewire **102**. For example, after a distal end of the guidewire **102** reaches the target location in the patient, the practitioner may twist the collar **170** in a first predetermined direction which clamps the collet **180** (not shown in FIG. 2B) onto the guidewire **102**. Twisting the collar **170** in the first predetermined direction axially displaces the collar **170** relative to the handle **160** which provides the compressing force to the collet **180**. The amount of twisting can affect the amount of compressive force. When the guidewire **102** is clamped and secured by the guidewire clamping device **150**, the guidewire **102** cannot axially slide through the guidewire clamping device **150**. When the guidewire **102** is clamped and secured by the guidewire clamping device **150**, the practitioner may grip the handle **160** and rotate the guidewire clamping device **150** and simultaneously rotate the guidewire **102**, which also rotates the distal tip of the guidewire **102**. The practitioner may also axially adjust the position of the guidewire **102** when the guidewire **102** is clamped and secured by the guidewire clamping device **150**. The practitioner may advance or retract the relative position of the guidewire clamping device **150** which simultaneously advances or retracts the axial position of the guidewire **102**.

[0038] The practitioner may release the guidewire **102** from the guidewire clamping device **150** by untwisting the collar **170** in a second predetermined direction, e.g., the opposite direction of the first predetermined direction, to unclamp the guidewire **102** from the guidewire clamping device **150**. Twisting the collar **170** in the second predetermined direction axially displaces the collar **170** relative to the handle **160** which causes the collet **180** to release the guidewire **102**. When the guidewire **102** is released and unclamped from the guidewire clamping device **150**, the practitioner may again axially advance or retract the guidewire **102** relative to the guidewire clamping device **150**.

[0039] FIG. 3 illustrates an exploded view of the guidewire clamping device **150** illustrated in FIG. 1. The guidewire clamping device **150** includes the handle **160**, the collar **170**, and the collet **180**. The handle **160** may couple to the collar **170** via the coupling member, such as a threaded interface **166**. The collet **180** may be partially disposed within both the handle **160** and the collar **170**. For example, a distal end **182** of the collet **180** may be housed within a lumen **173** the collar **170**, and a proximal end **181** of the collet **180** may be housed within a lumen **163** the handle **160**. In the illustrated embodiment, the distal end **182** of the collet **180** has a bulbous end that does not fit

within the lumen **163** of the handle **160** but does not fit within the lumen **173** of the collar **170**. Because the collet **180** does not entirely fit within the lumen **163** of the handle **160** the bulbous distal end **182** never enters the lumen **163**. However, the lumen **173** of the collar is designed to accommodate the bulbous distal end **182** in the unconstrained configuration. As the collar **170** and the handle **160** couple together, the bulbous distal end is axially displaced toward the handle **160** without the bulbous distal end **182** ever entering the lumen **163** of the handle **160** due to its size. [0040] FIGS. **4A** and **4B** illustrate perspective views of the handle **160**. The handle **160** includes a proximal end **161** and a distal end **162** with the lumen **163** extending from the proximal end **161** to the distal end **162** of the handle **160**. FIG. **4A** illustrates a distal end perspective view and FIG. **4B** illustrates a proximal end perspective view. The distal end **162** of the lumen **163** is configured to receive the proximal end **181** of the collet **180**.

[0041] The handle **160** comprises a flange **164** disposed between the proximal end **161** and the distal end **162**. The proximal side of the flange **164** includes a gripping member **165** that may be gripped by the practitioner. The gripping member **165** may comprise a textured surface or a rough surface. The textured or rough surface increases the friction of the gripping member **165**, thus increasing the ability of the practitioner to grip the gripping member **165**. The distal end **162** of the handle **160** comprises the threaded interface **166** that is engageable with the collar **170**.

[0042] FIGS. **5A** and **5B** illustrate perspective views of the collar **170**. The collar **170** includes a proximal end **171** and a distal end **172** with the lumen **173** extending from the proximal end **171** to the distal end **172** of the collar **170**. The lumen **173** decreases in diameter from the proximal end **171** to the distal end **172**. FIG. **5A** illustrates a distal end perspective view and FIG. **5B** illustrates a proximal end perspective view. The proximal end **171** of the collar **170** is configured to receive the distal end **182** of the collet **180**. The distal end **172** of the lumen **173** includes an aperture **174**. The aperture **174** is sized to receive different sizes of guidewires commercially available, specifically up to a diameter of 0.046 inches.

[0043] An outer surface **175** of the collar **170** includes the plurality of gripping elements **176**. The plurality of gripping elements **176** helps the practitioner to grip the collar **170** in order to rotate the collar **170** to either clamp the guidewire **102** or release the guidewire **102**. To clamp the guidewire **102**, the practitioner rotates the collar **170** in the first predetermined direction and to release the guidewire **102**, the practitioner rotates the collar **170** in the second predetermined direction, e.g., the opposite direction. In some embodiments, the first predetermined direction is clockwise and the second predetermined direction is counterclockwise.

[0044] FIG. **5B** illustrates the proximal end perspective view. An inner surface **177** of the collar **170** includes a threaded interface **178** that engages with the threaded interface **166** of the handle **160** and a distal inner surface **179** that is configured to engage with an outer surface **194** of the collet **180**, as discussed below.

[0045] FIGS. **6A-6D** illustrate different views of the collet **180**. FIG. **6A** illustrates a proximal perspective view of the collet **180**. FIG. **6B** illustrates a side view of the collet **180**. FIG. **6C** illustrates a distal end view of the collet **180**. FIG. **6D** illustrates a distal end perspective view of the collet **180**.

[0046] As discussed previously, the collet **180** comprises a body **183** with the proximal end **181** and the distal end **182**. The proximal end **181** of the body **183** includes a tubular section **184** with a lumen **185**. The tubular section **184** of the collet **180** is insertable into the lumen **163** of the handle **160**.

[0047] The tubular section **184** of the body **183** transitions into a plurality ramps **186**. The plurality of ramps **186** are separated circumferentially by a plurality of gaps **187**. In other words, a gap **187** is disposed between each of the adjacent ramps **186**. The plurality of ramps **186** flare radially outward from the tubular section **184**. In the illustrated embodiment, the collet **180** comprises four ramps **186**. However, the present disclosure is not so limited, and the collet **180** may have more or less than four ramps **186**.

[0048] FIG. 6B illustrates a side view of the collet **180**. In an unconstrained state, the plurality of ramps **186** flare radially outward from the tubular section **184** at a first predetermined angle θ_1 or draft angle. The first predetermined angle θ_1 may be between 2 degrees and 30 degrees. In some embodiments, the first predetermined angle θ_1 may be between 5 degrees and 10 degrees. Each ramp **186** comprises an elongate portion **188** and a bulbous distal portion **189**. The elongate portion **188** of each ramp **186** may be inserted into and constrained by the lumen **163** of the handle **160** and the bulbous distal portion **189** is inserted into the collar **170** such that the first predetermined angle θ_1 of the ramps **186** are flattened to second predetermined angle θ_2 of approximately zero degrees (see FIG. 7B). The bulbous distal portion **189** of each ramp **186** is sized so that it does not fit within the lumen **163** of the handle **160** and is configured to be disposed within the lumen **173** of the collar **170**. As discussed previously, twisting of the collar **170** onto the handle **160** is configured to further constrain the collet **180** from a partially constrained state to a constrained state in which the ramps **186** are biased inward to from the second predetermined angle θ_2 up to a third predetermined angle θ_3 . The twisting of the collar **170** is configured to secure the guidewire **102** with the collet **180**. The third predetermined angle θ_3 is less than second predetermined angle θ_2 and the third predetermined angle θ_3 may be approximately zero degrees. In some embodiments, the second predetermined angle θ_2 and the third predetermined angle θ_3 may be negative. The ramps **186** can be constrained anywhere between the first predetermined angle θ_1 and the third predetermined angle θ_3 . While the ramps **186** are at least partially constrained the elongate portion **188** of the ramps **186** may slide into the lumen **164** of the handle **160**.

[0049] Each of the plurality of ramps **186** comprises a contact surface **190** on an inner portion of the ramp **186**. The contact surfaces **190** are illustrated in FIGS. 6C and 6D. The contact surface **190** of each ramp **186** is configured to engage with the guidewire **102** when the collet **180** transitions from the unconstrained state to the constrained state. The contact surface **190** of each ramp **186** may be flat or slightly curved, e.g., concave curve. In other words, and in contrast to designs wherein the gripping portion of a collet come to a point or apex, the contact surface **190** of the collet **180** may be configured to engage a guidewire along a surface (**190**) defining a width, rather than only along an apex, point, or line. A point, apex, or sharp edge of other designs could cut through or puncture coatings on some guidewires or other devices. This can be especially problematic when the collet is brass or another metal as it can more easily cut through the coating of the guidewire **102**. The collet **180** of the present design comprises contact surfaces **190** that may distribute the gripping load and avoid cutting or otherwise damaging the wire surface.

[0050] As illustrated in FIGS. 6D, the contact surface **190** of each ramp **186** extends a majority of a length of each ramp **186** in a longitudinal direction of the collet **180**, including the elongate portion **188** and the bulbous distal portion **189**.

[0051] In some embodiments, the contact surface **190** of each ramp **186** may comprise a change in slope in the longitudinal direction of the collet **180**. In the illustrated embodiment of FIGS. 6C and 6D, the contact surface **190** may comprise a first sloped surface **191**, a second sloped surface **192**, and a transition point **193** between the first sloped surface **191** and the second sloped surface **192**. In certain circumstances, when the collet **180** is constrained, a distal portion of the second sloped surface **192** is configured to engage with the guidewire **102** before the first sloped surface **191** or the transition point **193**. In other words, the second sloped surface **192** of each contact surface engages the guidewire **102** before the first sloped surface **191** or the transition point **193**.

[0052] In some embodiments, when the collet **180** is constrained, the transition point **193** is configured to engage with the guidewire **102** before the first sloped surface **191** or the second sloped surface **192**. In other words, the transition point **193** of each contact surface engages the guidewire **102** before the contact surface **190** on opposing sides of the transition point **193**.

[0053] In some embodiments, when the collet **180** is constrained, a portion of the first sloped surface **191** is configured to engage with the guidewire **102** before the second sloped surface **192** or the transition point **193**. In other words, the first sloped surface **191** of each contact surface engages

the guidewire **102** before the second sloped surface **192** or the transition point **193**.
[0054] In some embodiments, the collet **180** is fabricated from a compliant material, such as a plastic, not a metal or a metal alloy. The term compliant signifies that the material may compress. For example, when the contact surface **190** of the collet **180** engages with the guidewire **102** when the collet **180** is being constrained, the collet **180** may compress. The compression of the collet **180** may increase the surface area of the contact surface **190** that engages with the guidewire **102** when the collet **180** is constrained. The compression of the collet **180** also prevents excessive force being applied to the guidewire **102** and prevent the potential cutting of the coating of the guidewire **102**. Exemplary compliant materials include plastics such as polyoxymethylene (e.g., Delrin®), nylon, polycarbonate, polybutylene terephthalate, acrylonitrile butadiene styrene, and the like, and any suitable combination thereof.

[0055] As discussed above, in some embodiments, the contact surface **190** includes the first sloped surface **191**, the second sloped surface **192**, and the transition point **193**. When the collet **180** is fabricated from a compliant material, the transition point **193** is configured to engage with the guidewire **102** before the first sloped surface **191** or the second sloped surface **192**. Due to the compliant nature of the collet **180**, the transition point **193** will compress and then the first sloped surface **191** and the second sloped surface **192** will engage with the guidewire **102** and increase the surface area of the engagement between the contact surface **190** and the guidewire **102**. In some circumstances the contact surface **190** conforms to the shape of the guidewire **102**.

[0056] The guidewire clamping device **150** is configured to accommodate a variety of different sized guidewires. Specifically, the collet **180** is design to accommodate guidewires with a diameter that range between 0.009 inches and 0.046 inches. The various features of the collet **180** enable the collet **180** to accommodate a wide variety of different size diameter guidewires. Specifically, the flared design of the ramps **186** enables the collet **180** to accommodate larger diameter guidewires, up to 0.046 inches. The design of the ramps **186** of the collet **180** also enables the collet **180** to accommodate smaller diameter guidewires, down to 0.009 inches. Prior art collets require different sized collets to accommodate the same range of available guidewires, a smaller collets to accommodate smaller guidewires and larger collets to accommodate larger guidewires.

[0057] In some embodiments, the outer surface **194** of the bulbous distal portion **189** comprises a slot **195**. The outer surface **194** is configured to engage with the distal inner surface **179** of the collar **170**. The slot **195** disposed on the outer surface **194** of the bulbous distal portion **189** of the plurality of ramps **186** make the collet **180** light and avoids voids in the collet **180** during the molding process.

[0058] FIGS. 7A-7C illustrate the effects of an adjustment to the threaded interface **166** of the handle **160** and an adjustment to the threaded interface **178** of the collar **170**.

[0059] FIG. 7A illustrates a cross-sectional view of the guidewire clamping device **150** of FIG. 1 taken through plane 7A-7A in a first position where the guidewire **102** is unsecured and may be manipulated in an axial or rotational direction. As shown, a diameter of the lumen **185** within the collet **180** may be large enough to allow the guidewire **102** to pass through. In this first position, the practitioner may push or pull the guidewire **102** freely to adjust its longitudinal position. For example, when a practitioner places the guidewire **102** in a patient, the practitioner may advance the guidewire **102** to a treatment site while the guidewire clamping device **150** is in this first position.

[0060] FIG. 7B illustrates a cross-sectional view analogous to the view of FIG. 7A of the guidewire clamping device **150** of FIG. 1 in a second position or a partially constrained state. To secure the guidewire **102**, the collar **170** is coupled to the handle **160**. In the illustrated embodiment, threaded interface **178** of the collar **170** engages with the threaded interface **166** of the handle **160**. As the practitioner inserts the handle **160** and the collet **180** into the collar **170**, the distal inner surface **179** of the collar **170** engages with the outer surface **194** of the collet **180** due to axial displacement between the collar **170** and the handle **160**. This engagement forces the collet **180** into the lumen

163 of the handle **160**, which biases the plurality of ramps **186** inward and constrains the collet **180**. The plurality of ramps **186** are constrained from the first predetermined angle θ_1 toward the second predetermined angle θ_2 .

[0061] In the second position, where the guidewire **102** is secured by the contact surface **190** of each of the plurality of ramps **186** of the collet **180**. As illustrated in the FIG. 7B, the distal end of the second contact surface **192** engages with the guidewire **102**. The engagement between the contact surface **190** of the plurality of ramps **186** of the collet **180** clamps onto the guidewire **102** and secures the guidewire **102** in place. As discussed previously, when the guidewire **102** is secured, the practitioner may rotate the handle **160**, which simultaneously rotates the guidewire **102**.

[0062] FIG. 7C illustrates a cross-sectional view analogous to the view of FIGS. 7A and 7B of the guidewire clamping device **150** of FIG. 1 in a third position or a fully constrained state. To further secure the guidewire **102**, the threaded interface **178** of the collar **170** engages with the threaded interface **166** of the handle **160** is twisted in the first predetermined direction which causes additional axial displacement. As the practitioner twists the collar **170** in the first predetermined direction, the distal inner surface **179** of the collar **170** further engages with the outer surface **194** of the collet **180**. This engagement forces the contact surface **190** of each of the plurality of ramps **186** to further engage with the guidewire **102**. The collet **180** is fabricated from a compliant material. In other words, the collet **180** compresses and conforms to the shape of the guidewire **102**. In the third position, the plurality of ramps **186** are constrained to the third predetermined angle θ_3 .

[0063] In the third position, where the guidewire **102** is secured by the contact surface **190** of each of the plurality of ramps **186** of the collet **180**. In some embodiments, only the second contact surface **192** of each of the plurality of ramps **186** engages with the guidewire **102**. In some embodiments, the second contact surface **192** and the transition point **193** of each of the plurality of ramps **186** engages with the guidewire **102**. In some embodiments, the entire contact surface **190** engages with the guidewire **102**, including the first sloped surface **191**, the second sloped surface **192**, and the transition point **193**. The engagement between the contact surface **190** of the plurality of ramps **186** of the collet **180** clamps onto the guidewire **102** and secures the guidewire **102** in place. As discussed previously, when the guidewire **102** is secured, the practitioner may rotate the handle **160**, which simultaneously rotates the guidewire **102**.

[0064] Any methods disclosed herein include one or more steps or actions for performing the described method. The method steps and/or actions may be interchanged with one another. In other words, unless a specific order of steps or actions is required for proper operation of the embodiment, the order and/or use of specific steps and/or actions may be modified. Moreover, sub-routines or only a portion of a method described herein may be a separate method within the scope of this disclosure. Stated otherwise, some methods may include only a portion of the steps described in a more detailed method.

[0065] Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure, or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

[0066] Similarly, it should be appreciated by one of skill in the art with the benefit of this disclosure that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim requires more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with

their dependent claims.

[0067] Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the present disclosure.

Claims

1. A collet for a guidewire clamping device, the collet comprising: a body comprising: a tubular section with a lumen; a plurality of ramps that are flared outward from the tubular section in a unconstrained state, wherein in a constrained state, the plurality of ramps are biased inward such that a contact surface of each of the ramps are configured to contact a guidewire disposed within the body and secure the guidewire when all of the contact surface of each ramp engages the guidewire.
2. The collet of claim 1, wherein the plurality of ramps are flared outward at a draft angle between 2 degrees and 30 degrees.
3. The collet of claim 1, wherein the contact surface of each ramp of the plurality of ramps is not a point.
4. The collet of claim 1, wherein the contact surface of each ramp of the plurality of ramps is flat.
5. The collet of claim 1, wherein the contact surface of each ramp of the plurality of ramps extends a majority of a length of the ramp in a longitudinal direction of the collet.
6. The collet of claim 1, wherein the contact surface of each ramp of the plurality of ramps comprises a change in slope in a longitudinal direction of the collet.
7. The collet of claim 1, wherein each ramp comprises a slot disposed on an outer surface of the ramp.
8. The collet of claim 1, wherein the collet comprises four ramps.
9. The collet of claim 1, wherein the collet is fabricated from a compliant material such that when the plurality of ramps clamp onto the guidewire, the compliant material is configured to compress.
10. The collet of claim 1, wherein the collet is a plastic.
11. The collet of claim 10, wherein the collet is polyoxymethylene.
12. A guidewire clamping device system comprising: a handle comprising a body that defines a lumen that extends from a proximal end of the handle to a distal end of the handle, the distal end of the handle comprising a coupling member; a collar that defines a lumen that extends from a proximal end to a distal end of the collar, wherein the collar is configured to couple to the distal end of the handle; and a collet comprising: a body comprising: a tubular section with a lumen; and a plurality of ramps that are flared outward from the tubular section in a unconstrained state, wherein the collet is disposed within a lumen of the handle and the collar, and wherein the collar is coupled to the handle, the collar biases the plurality of ramps of the collet inward to a constrained state to constrain the collet.
13. The guidewire clamping device system of claim 12, further comprising a guidewire that extends through the handle, the collet, and the collar, wherein in the constrained state, the plurality of ramps are biased inward by the collar such that a contact surface of each of the ramps are configured to contact the guidewire and secure the guidewire when all of the contact surface of each ramp engages the guidewire.
14. The guidewire clamping device system of claim 13, wherein the guidewire comprises a diameter between 0.009 inches and 0.046 inches and the collet is configured to secure guidewires with a diameter between 0.009 inches and 0.046 inches.
15. The guidewire clamping device system of claim 13, wherein the contact surface of each ramp of the plurality of ramps comprises a change in slope in a longitudinal direction of the collet, and wherein the change in slope and the contact surface distal to the change in slope engages the

guidewire.

16. The guidewire clamping device system of claim 15, wherein the collet is fabricated from a compliant plastic material, and wherein the plurality of ramps of the collet compress when the contact surfaces engage with the guidewire.

17. A method of clamping a guidewire comprising: inserting a guidewire into a guidewire clamping device, the guidewire clamping device comprising: a handle comprising a body that defines a lumen that extends from a proximal end of the handle to a distal end of the handle, the distal end of the handle comprising a coupling member; a collar that defines a lumen that extends from a proximal end to a distal end of the collar, wherein the collar is configured to couple to the distal end of the handle; and a collet comprising: a body comprising: a tubular section with a lumen; a plurality of ramps that are flared outward from the tubular section in a unconstrained state; twisting the collar of the guidewire clamping device to constrain the plurality of ramps inward such that a contact surface of each of the plurality of ramps clamp the guidewire.

18. The method of claim 17, wherein the collet is fabricated from a compliant plastic material such that twisting the collar compresses the collet and the contact surface of each ramp conforms to the guidewire.

19. The method of claim 17, wherein the contact surface of each ramp of the plurality of ramps comprises a transition point that changes a slope of the contact surface, and wherein the transition point and the contact surface distal to the transition point of each contact surface engages the guidewire.

20. The method of claim 17, wherein the collet of the guidewire clamping device is configured to secure guidewires with diameters between 0.009 inches and 0.046 inches.
